The Rise of Algorithmic Retail Options Traders

Diego Amaya, Pedro A. Garcia-Ares, Neil D. Pearson, and Aurelio Vasquez

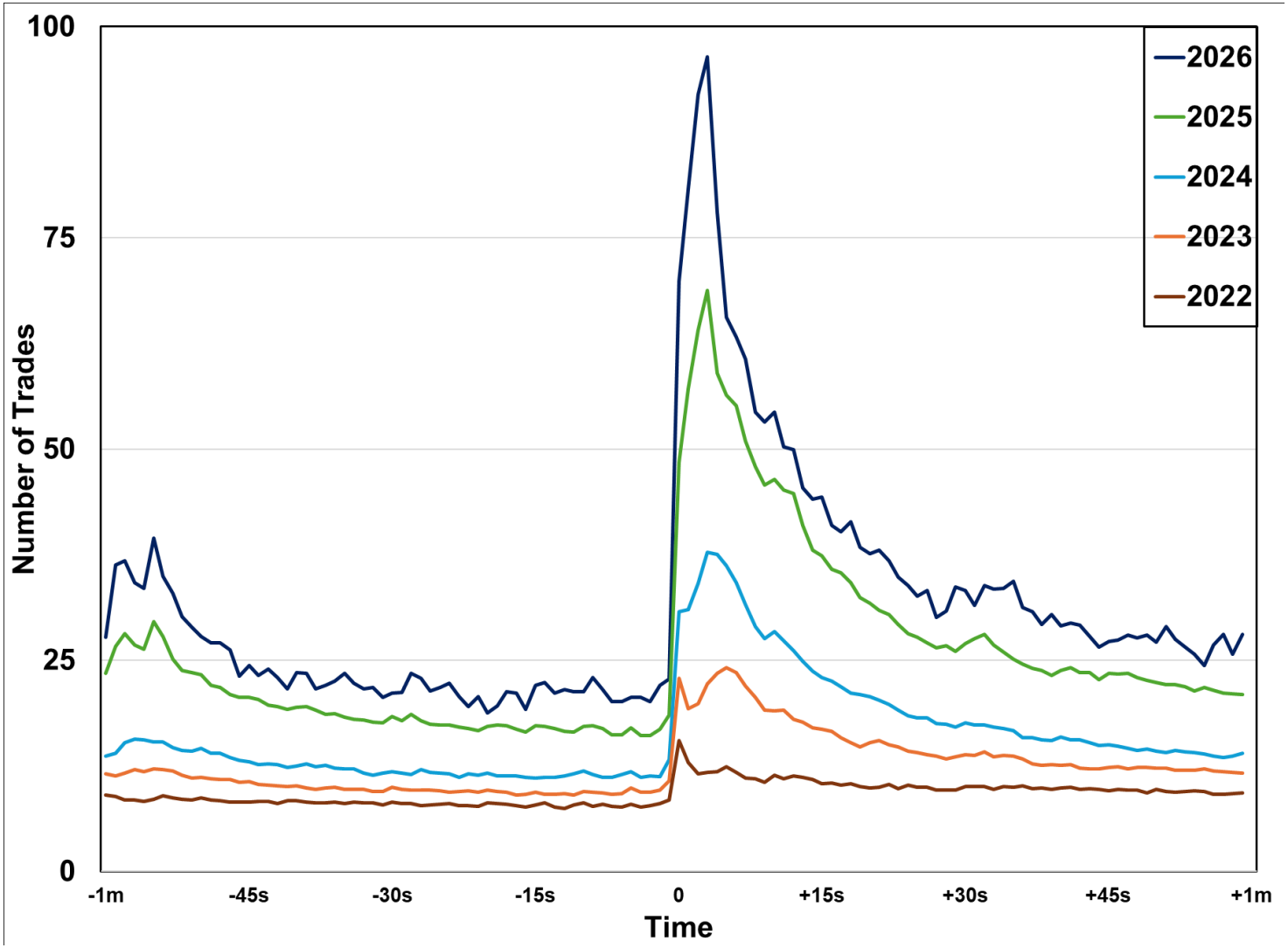
Discussion by Justin Mohr
University of Mississippi

FMA Annual Meeting
October 2026

Overview of Paper

- Examine SPX 0DTE option trades
  - OPRA transactions, May 2022 to January 2026
- Find trading spikes at fixed clock times
  - Concentrated in small, defined-risk spreads: short verticals and iron condors
  - Look like templates from retail automation platforms
- Show liquidity moves both ways
  - Quoted spreads narrow, effective spreads widen
- "Synchronization without coordination"

The Spikes


100
75
50
25
0
Number of Trades
-1m
-45s
-30s
-15s
0
+15s
+30s
+45s
+1m
Time
2026
2025
2024
2023
2022

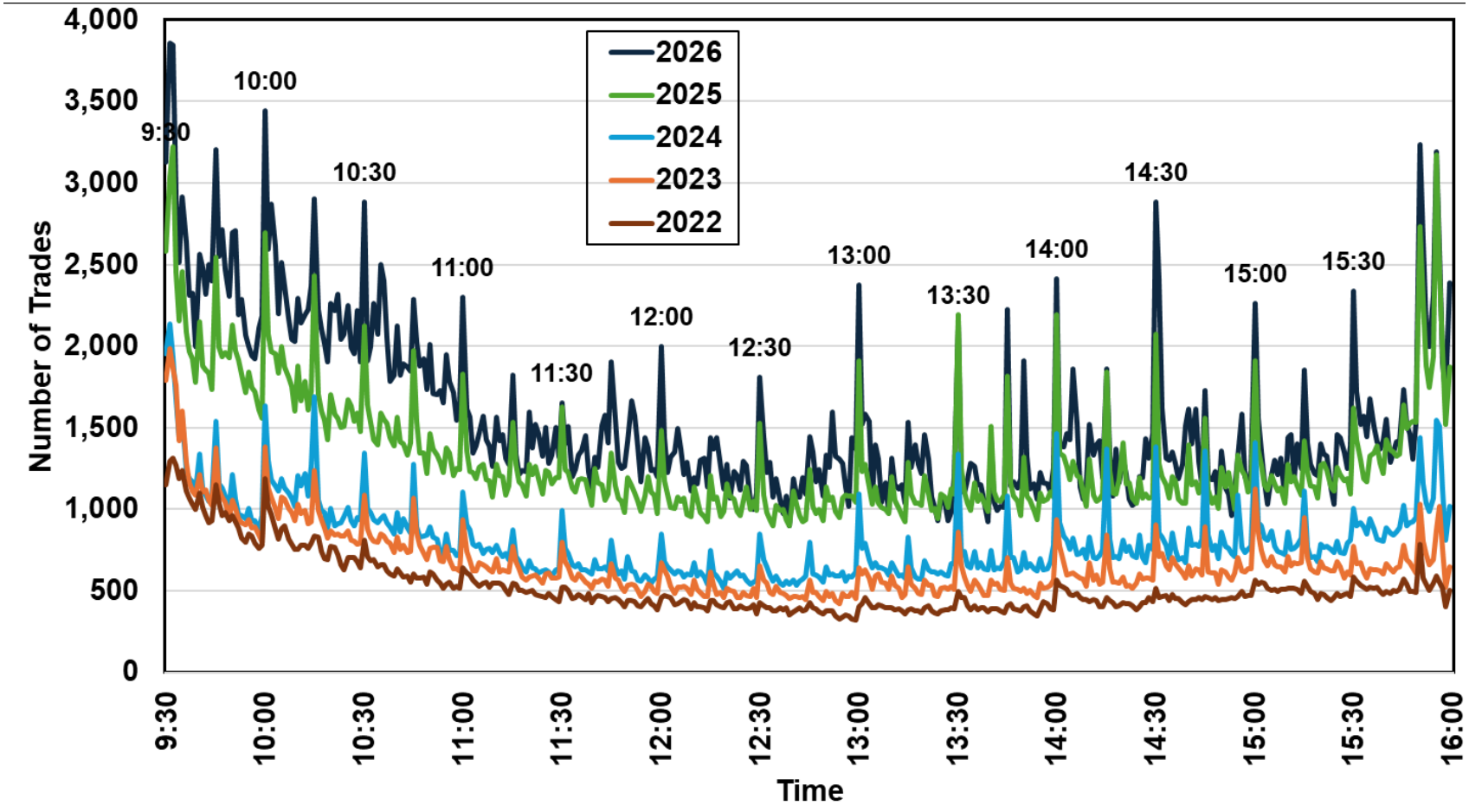
Focus of Comments

- Bots do not learn, communicate, or interact
  - Convincing!
- But what about everyone else?
  - Market makers
  - Informed traders
- My comments are all a version of this

Comment 1: How long to learn?

- Quoted spreads narrow at the spikes
  - Market makers see it coming
- But did they always?
  - Small, new spike: market maker does not know what it is
  - As the spike grows, we should see the narrowing
- New spikes turn on during the sample (possible retail adjustment?)
  - 2026 quarter-hour spikes (11:45, 13:45, 15:15) are as large as 2025 half-hour spikes
- Spread results pool 2024 to 2026

Comment 1: How long to learn?



Comment 1: How long to learn?

- Quoted spreads narrow at the spikes
  - Market makers see it coming
- But did they always?
  - Small, new spike: market maker does not know what it is
  - As the spike grows, we should see the narrowing
- New spikes turn on during the sample (possible retail adjustment?)
  - 2026 quarter-hour spikes (11:45, 13:45, 15:15) are as large as 2025 half-hour spikes
- Spread results pool 2024 to 2026
- Suggestion:
  - Test the spreads by year
  - Speed of learning?

Comment 2: Why not hide?

- "The cost is not adverse selection"
  - For the bots, sure
- But who else trades in that minute?
  - Classifier finds 61% of burst trades
  - Single-leg trades spike too
- Predictable noise is a great place to hide

Comment 2: Why not hide?

- "The cost is not adverse selection"
  - For the bots, sure
- But who else trades in that minute?
  - Classifier finds 61% of burst trades
  - Single-leg trades spike too
- Predictable noise is a great place to hide
- Suggestion:
  - Table 9 for the single-leg trades
  - More price impact in the spike minute?

Comment 3: Announcement days?

- FOMC at 14:00 is special
  - "We treat that window separately"
  - But (I think...) still in the pooled results
- What about 10:00 macro releases?
  - Both spreads widen at 10:00
- Only windows where quoted spreads do not narrow: 10:00 and 14:00

Comment 3: Announcement days?

Change at the spike minute vs. preceding five minutes ($), Table IA2

	10:00	14:00
Short call vertical		
Quoted spread	0.0187 (6.1)	−0.0001 (0.0)
Effective spread	0.0092 (10.5)	0.0073 (7.5)
Realized spread, 10 min	−0.0100 (−0.3)	0.0383 (1.2)
Short put vertical		
Quoted spread	0.0283 (4.6)	0.0033 (0.7)
Effective spread	0.0084 (9.4)	0.0066 (6.7)
Realized spread, 10 min	−0.0054 (−0.2)	0.0033 (0.1)
Short iron condor		
Quoted spread	0.0308 (5.6)	0.0201 (4.3)
Effective spread	0.0136 (12.7)	0.0111 (8.9)
Realized spread, 10 min	−0.0138 (−0.7)	0.0241 (1.2)

Comment 3: Announcement days?

- FOMC at 14:00 is special
  - "We treat that window separately"
  - But (I think...) still in the pooled results
- What about 10:00 macro releases?
  - Both spreads widen at 10:00
- Only windows where quoted spreads do not narrow: 10:00 and 14:00
- Without announcement days, do these windows look like the rest?
- Suggestion:
  - Examine announcement days separately

Conclusion

- Very interesting paper, a clean look at something new in retail trading
- The bots do not learn, but I would like to see who does
- Overall, enjoyable read. Good luck to the authors!